



FIELD TRIAL · L. VANNAMEI · DEC 2025 – MAY 2026

Probiotic Pond Performance — Analysis Report

2J607 (OBT) was the **standout pond** — top biomass **5,437 kg** (+4.7%), best **FCR 1.67**, highest **survival 58.5%** and **+10% profit** vs control. Yet across the four OBT ponds the group *averaged below* control (biomass -8.7%, survival -13.9%, FCR +11.1% worse): the probiotic effect was **inconsistent**, and **survival** — not growth or water quality — decided every outcome.

Control **2D407** vs four OBT ponds **2E707 · 2E807 · 2J407 · 2J607** · identical 505,000 PL stocking · 136–145 DOC · live NSA Final Analysis Sheet. Eight analysis sections, each point as its own card or interactive graph.

BEST POND

2J607

OBT · wins every KPI

TOP BIOMASS

5,437 kg

+4.7% vs control

BEST FCR

1.67

control 1.71

TOP SURVIVAL

58.5%

vs 43.9% worst

TOP PROFIT

\$14.7k

+10% vs control

PROBIOTIC VERDICT

Inconsistent

1 of 4 OBT beat control

The intelligence in one line

On **average** the four treated ponds **trailed** the control (biomass -8.7%, survival -13.9%, FCR +11.1% worse) — yet the single best pond, **2J607 (OBT)**, beat the control on every metric and was the only one more profitable (+10%). High **variance within the treatment**, not a clean win/loss. The master lever throughout was **survival**.

Comparative Pond Analysis

Treated ponds vs the control across survival, biomass, ABW, growth, feed, FCR, productivity & harvest. All ponds stocked the same 505,000 PL, so these are pure outcome differences. Absolute and % differences shown.

1.1 Harvest scorecard (absolute values, best→worst shaded)

★ Ask AI — explain this chart

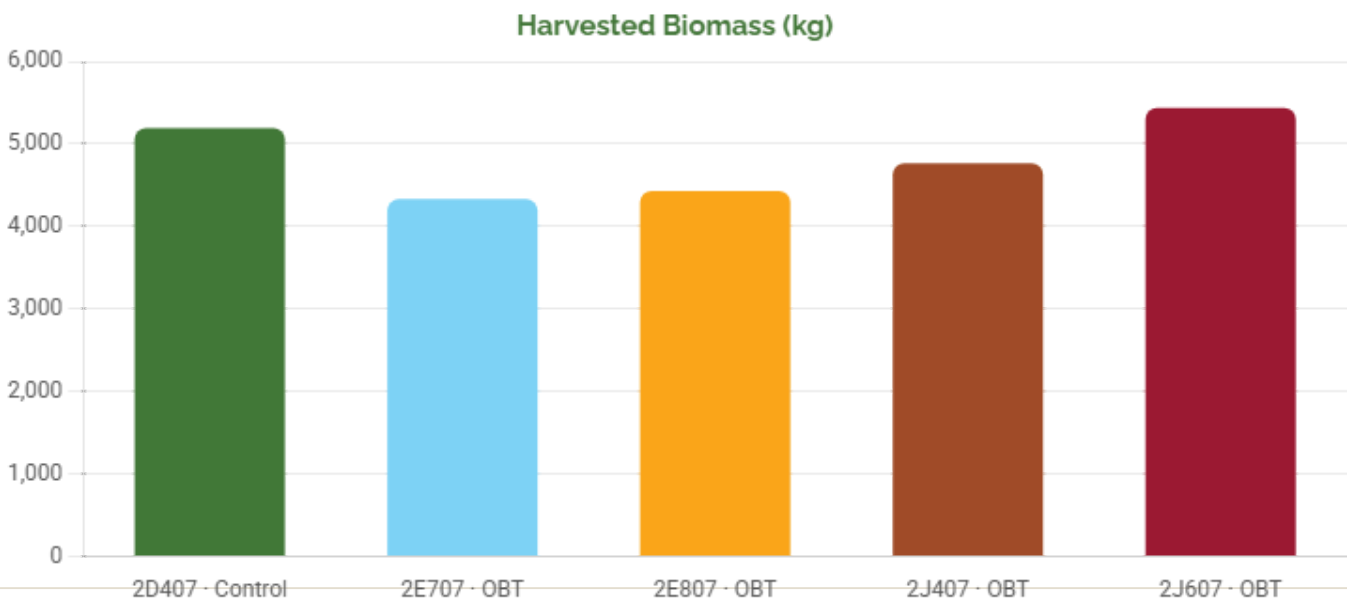
Metric	2D407 Control	2E707 OBT	2E807 OBT	2J407 OBT	2J607 OBT
Biomass (kg)	5,191	4,330	4,427	4,763	5,437
ABW (g)	17.71	16.45	19.96	20.82	18.39
FCR	1.71	1.99	1.95	1.99	1.67
Survival %	58.0	52.1	43.9	45.3	58.5
Density (n/m²)	58	52	44	45	59
Profit (\$)	\$13,400	\$9,143	\$9,651	\$10,487	\$14,697
vs Control	base	-32%	-28%	-22%	+10%

Green = best, amber = middle, red = worst within each metric. FCR inverted (lower is better). Best value ringed.

1.2 Head-to-head — absolute values

- Biomass
- Survival
- ABW
- FCR
- Growth/day
- Feed used
- Productivity
- Profit

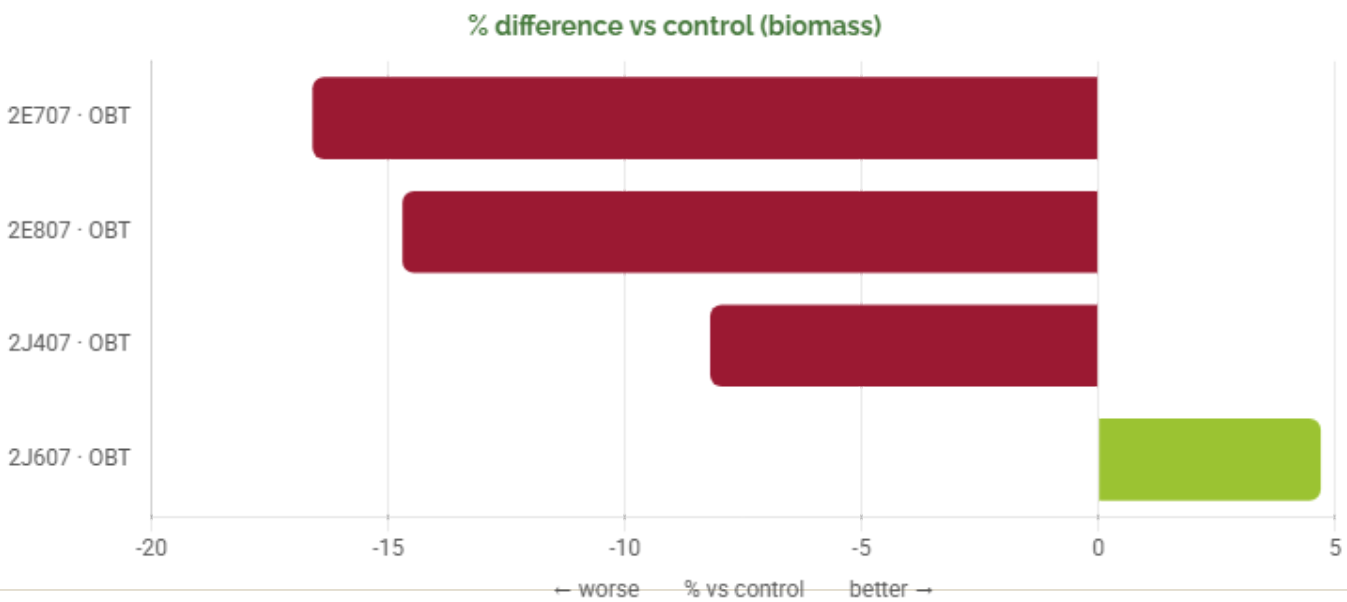
🌟 Ask AI — explain this chart



1.3 % difference vs control

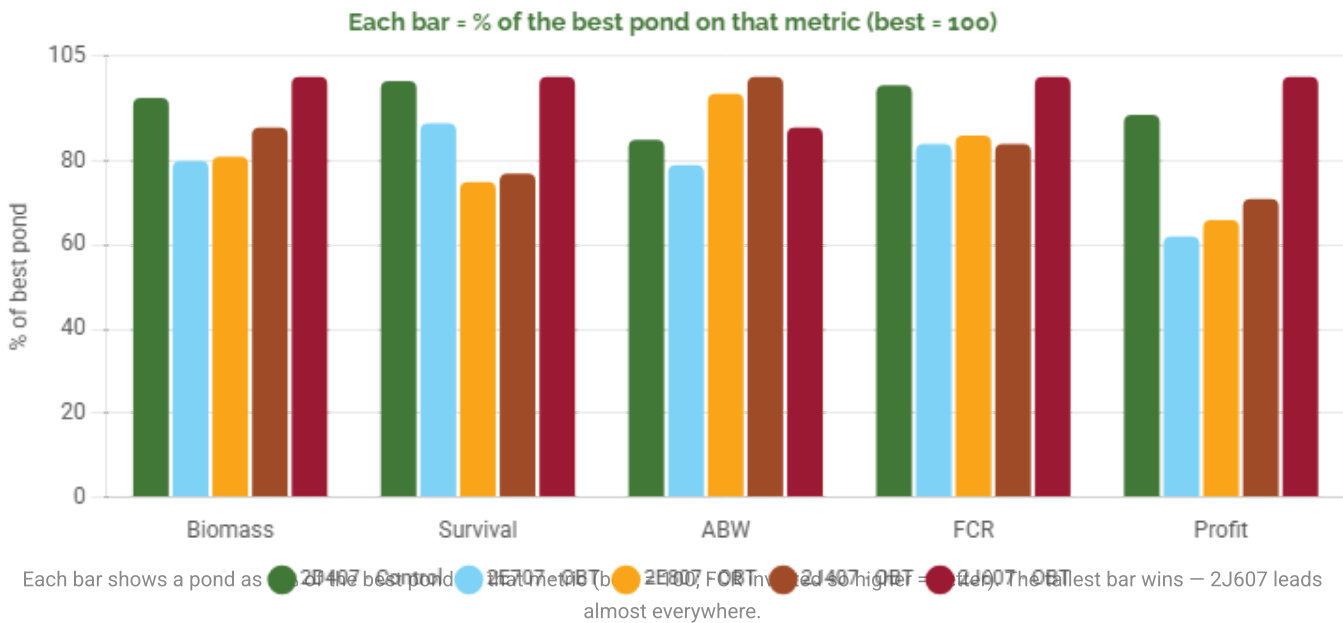
- Biomass
- Survival
- ABW
- FCR
- Profit
- Feed

🌟 Ask AI — explain this chart



1.4 How each pond scores on every metric

✦ Ask AI — explain this chart



1.5 Productivity & feed consumed

2D407 · CONTROL

38.2 kg/day

Feed 8,877 kg · 292,900 survivors · ADG 130.2 mg/d

2E707 · OBT

31.4 kg/day

Feed 8,617 kg · 263,105 survivors · ADG 119.2 mg/d

2E807 · OBT

32.3 kg/day

Feed 8,633 kg · 221,695 survivors · ADG 145.7 mg/d

2J407 · OBT

32.8 kg/day

Feed 9,478 kg · 228,765 survivors · ADG 143.6 mg/d

2J607 · OBT

40 kg/day

Feed 9,080 kg · 295,425 survivors · ADG 135.2 mg/d

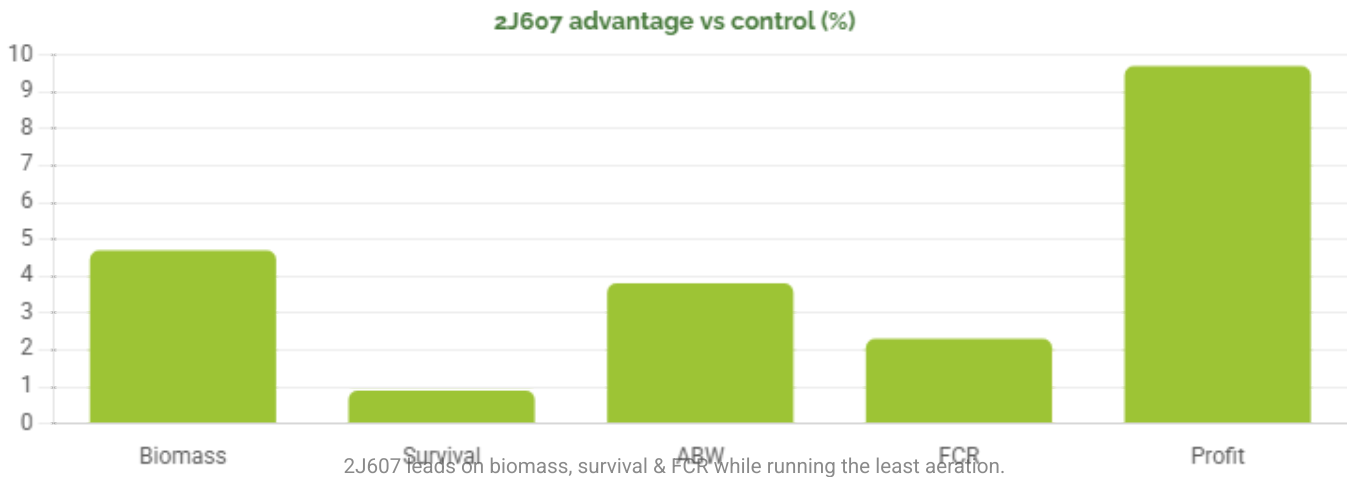
Productivity = harvested biomass ÷ days of culture. Feed used is total feed delivered per pond.

What Went Right

Positive outcomes, the evidence behind them, signs of a positive probiotic effect, and the business + technical implications.

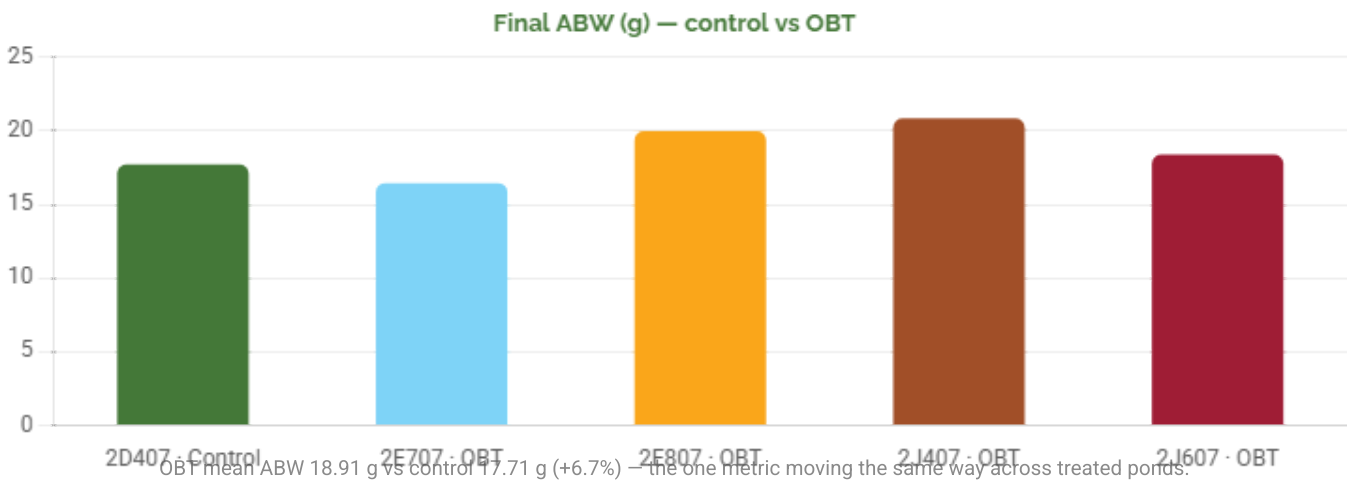
2.1 Best pond 2J607 — delta vs control

✦ Ask AI — explain this chart



2.2 Positive probiotic signal — growth (ABW)

✦ Ask AI — explain this chart



2.3 · 2J607 was the best pond in the trial

FINDING

An OBT pond topped every production metric and was the only one more profitable than control.

DATA

Biomass 5,437 kg, FCR 1.67, survival 58.5%, profit \$14,697 (+10%), lowest aeration (11.97 HP).

INTERPRETATION

It combined the best survival with above-average growth — neither side of biomass = survival × growth was sacrificed; a genuinely positive outcome.

STATISTICAL SUPPORT

Best FCR and best survival coincide in 2J607; across 5 ponds FCR tracks pH/afternoon-DO ($r \approx +0.9$, confounded by biomass load). $n=5$, directional.

RECOMMENDATION

Adopt 2J607 as the reference protocol; document its dosing & aeration and replicate before firm conclusions.

2.4 · Parameters that improved & the probiotic signal

FINDING

Treated ponds grew larger shrimp on average; WQ stability was maintained throughout.

DATA

OBT mean ABW 18.91 g vs control 17.71 g (+6.7%); pH 8.0–8.1, alkalinity 135–144 ppm, TAN/NO₂ near zero (except 2J407).

INTERPRETATION

A modest, consistent growth/feed-conversion edge across treated ponds is the clearest pro-probiotic signal; good husbandry kept chemistry from limiting any pond.

STATISTICAL SUPPORT

ABW also correlates with afternoon temp ($r = +0.88$), so part of the edge may be thermal — isolate next cycle at matched survival.

RECOMMENDATION

Keep fortnightly ABW logging; design a matched-survival comparison to confirm a true growth effect.

Technical implication

2J607 shows the protocol's **upside is real**: best survival + above-average growth simultaneously, plus the lowest aeration load (11.97 HP vs 12.85 control) → ~\$237 electricity saving. Replicating its dosing and aeration profile is the technical priority.

Business implication

2J607 returned **+10% profit vs control (\$14,697 vs \$13,400)** on near-identical cost. If reproducible across ponds, the gain scales directly with yield — but on current evidence (1 of 4) it is a **promising lead, not a bankable result**.

What Went Wrong

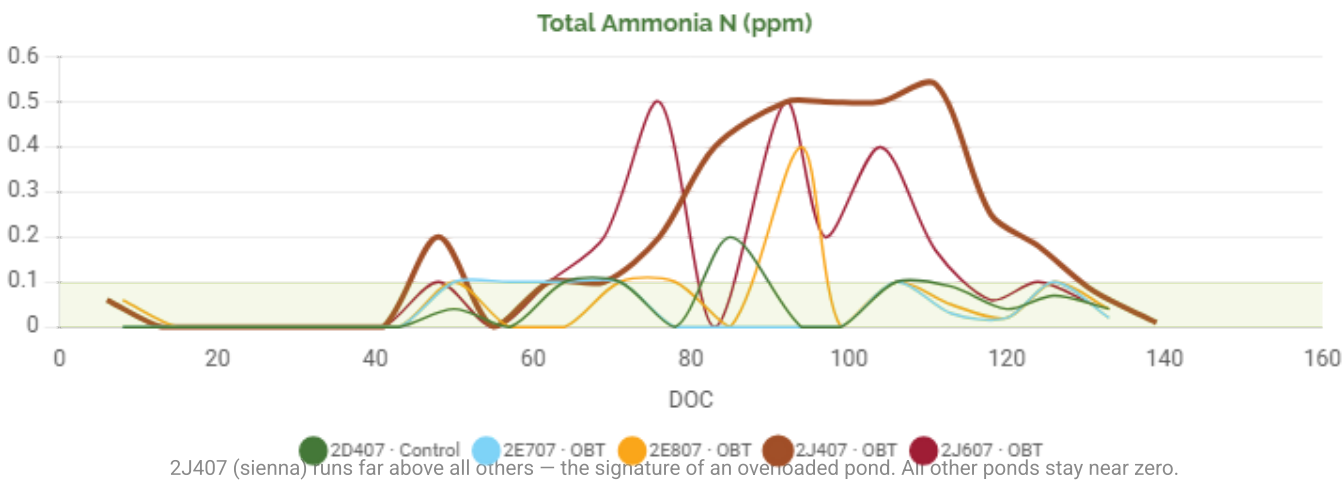
Underperformers, water-quality issues, FCR inefficiency, survival/growth shortfalls, and the root cause of each – with supporting evidence.

3.1 Water-quality issue — ammonia (TAN) anomaly

TAN (ammonia)

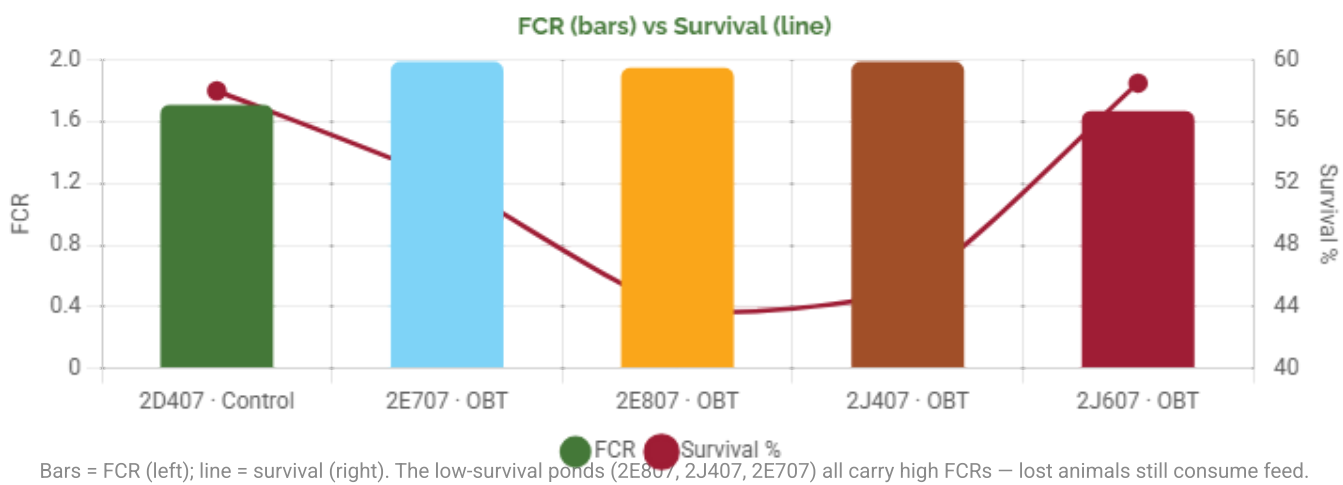
Nitrite NO₂

✦ Ask AI — explain this chart



3.2 FCR inefficiency tracks lost survival

✦ Ask AI — explain this chart



3.3 · Three of four OBT ponds lagged the control

FINDING

2E707, 2E807 and 2J407 each returned 22–32% less profit than control.

DATA

Biomass 4,330 / 4,427 / 4,763 kg (–16.6% / –14.7% / –8.2%); FCR 1.99 / 1.95 / 1.99; survival 52.1% / 43.9% / 45.3%.

INTERPRETATION

The treatment-group mean fell and variance rose — the probiotic effect is inconsistent and pond-dependent under current use.

STATISTICAL SUPPORT

OBT mean below control on 3/4 KPIs (biomass –8.7%, survival –13.9%, FCR +11.1%); n=4 treated, not conclusive but direction is unfavourable.

RECOMMENDATION

Do not scale OBT on current data; run a replicated randomised trial (≥4–5 ponds/arm).

3.4 · 2E807 & 2J407 — survival reduction despite top growth

FINDING

The two biggest-shrimp ponds had the two lowest survivals → weak biomass.

DATA

2E807: ABW 19.96 g, survival 43.9%, ~222k survivors. 2J407: ABW 20.82 g (highest), survival 45.3%, ~229k survivors. Control: ~293k.

INTERPRETATION

Heavy mortality thinned both ponds; survivors grew large on the freed feed/space. High ABW is a symptom of loss, not success.

STATISTICAL SUPPORT

Both are clear outliers on the survival-vs-ABW relationship; survival, not growth, capped their biomass.

RECOMMENDATION

Investigate early-to-mid mortality (seed batch, stocking stress, disease, DO dips); protect the first 60 DOC.

3.5 · 2J407 — nitrogen/WQ event, confirmed at harvest

FINDING

Highest ammonia & nitrite and lowest morning DO; harvested with the highest ABW but 2nd-lowest survival.

DATA

Mean TAN 0.19, NO₂ 0.06 ppm (3–6× peers); morning DO 4.81 ppm. Harvest: ABW 20.82 g, survival 45.3%, biomass 4,763 kg, FCR 1.99, –22% profit.

INTERPRETATION

Elevated TAN+NO₂ with depressed DO = an overloaded pond — and the harvest confirms it: heavy losses left fewer, larger shrimp and a mediocre yield.

STATISTICAL SUPPORT

2J407's TAN/NO₂ means are the dataset maxima; its survival is the 2nd-lowest, exactly matching the chemistry warning.

RECOMMENDATION

Set TAN>0.15 / NO₂>0.05 ppm as hard action triggers — cut feed, raise exchange/aeration early.

3.6 · Operational/environmental — late-culture DO squeeze

FINDING

Morning DO fell below ~4 ppm late in every pond.

DATA

~5.5 ppm early → 3.7–3.9 ppm last days; slope ~–0.015 ppm/day.

INTERPRETATION

Rising biomass + seasonal warming raise night oxygen demand; dawn DO sags near harvest, causing chronic sub-lethal stress.

STATISTICAL SUPPORT

Regression of morning DO on DOC negative & consistent ($r \approx -0.78$ to -0.80 , $p < 0.01$, $n \approx 136/\text{pond}$) — robust.

RECOMMENDATION

Add dawn-targeted aeration from ~DOC 90; consider capping late feed/biomass.

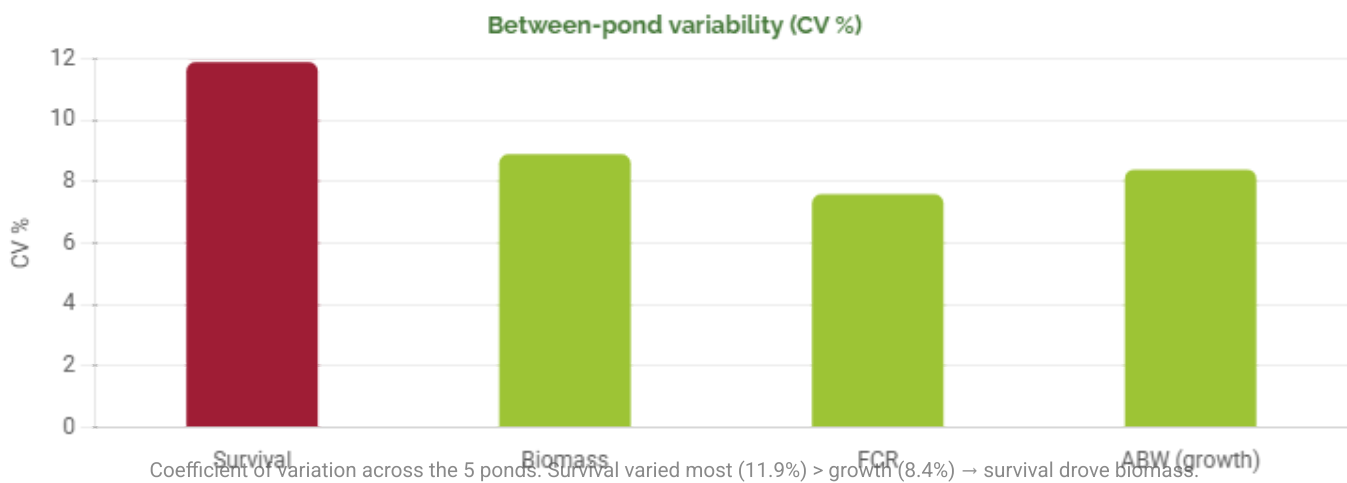
Update — pond 2J407 is now harvested (DOC 145): it carried the highest mean TAN (0.19 ppm) and nitrite (0.06 ppm) and the lowest morning DO of any pond — and finished with the second-lowest survival (45.3%) despite the highest ABW (20.82 g). That is exactly the overloaded-pond outcome its water chemistry warned of: heavy losses, big surviving shrimp, mediocre biomass (4,763 kg) and –22% profit.

Root-Cause Analysis

What most influenced biomass, survival and FCR — and whether differences were driven by survival, growth, water quality, feed efficiency or management.

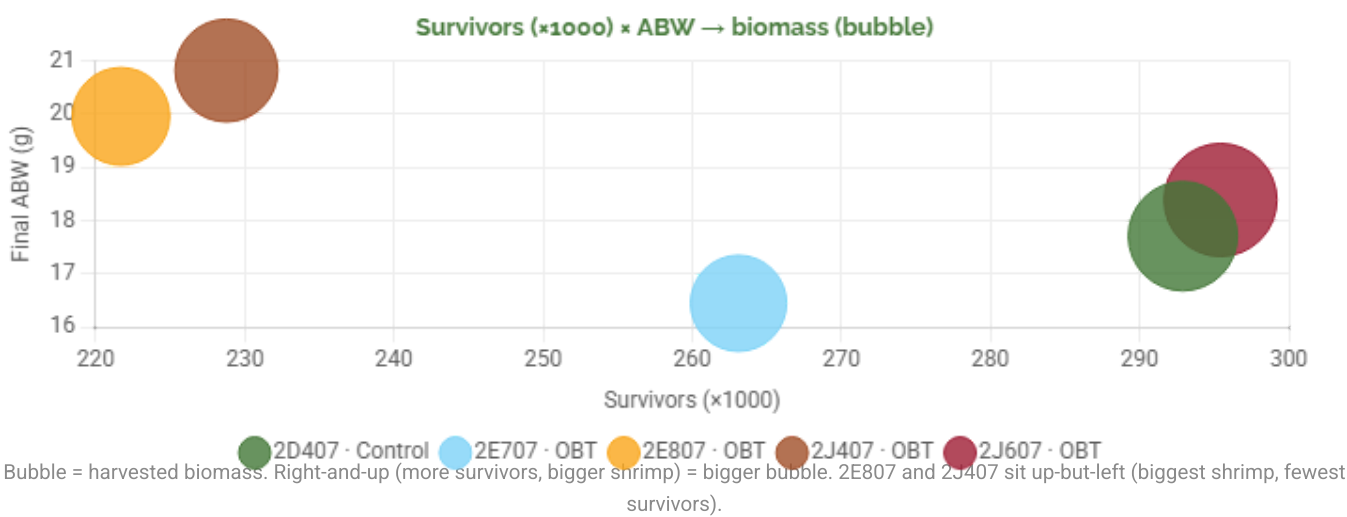
4.1 What drove the biomass spread? (variability)

✦ Ask AI — explain this chart



4.2 Biomass decomposition: survivors × ABW

✦ Ask AI — explain this chart



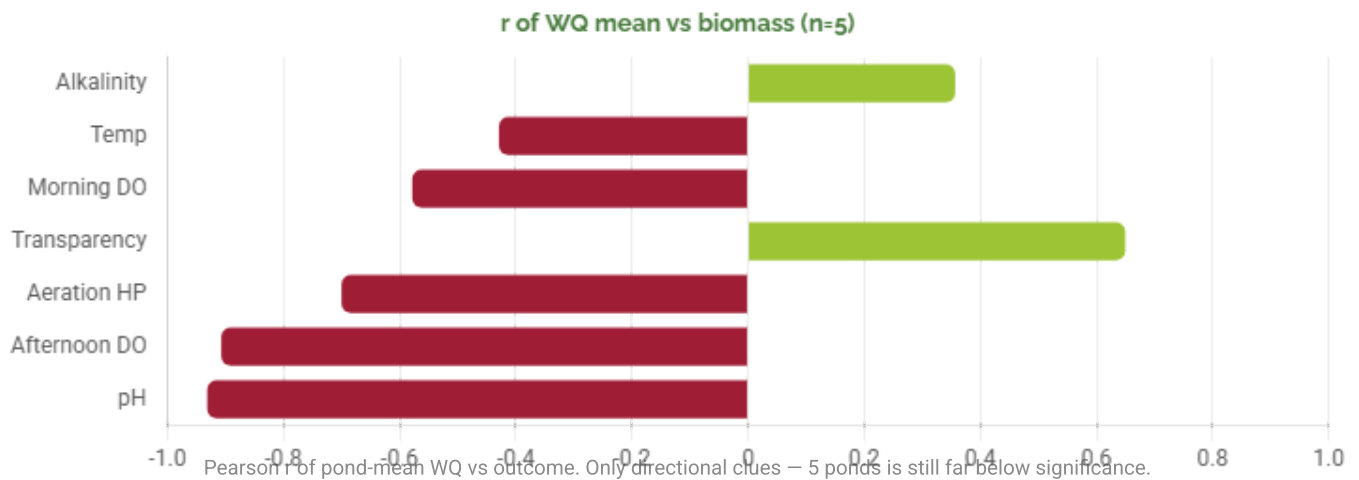
4.3 WQ ↔ performance correlations (n=5, directional)

Biomass

Survival

FCR

✦ Ask AI — explain this chart



4.4 What most influenced each outcome

BIOMASS

Survival

CV 11.9% > ABW 8.4%; top-2 biomass = top-2 survival

SURVIVAL

Health / seed

WQ near-identical yet survival ranges 14.6 pts; not routine chemistry

FCR

Survival

Best FCR where survival best; dead shrimp waste feed

Verdict: performance differences were driven primarily by **survival & pond-level health/management**, with routine water chemistry minor and the probiotic adding at most a small, inconsistent growth effect. Feed efficiency was an *outcome* of survival, not an independent cause.

Time-Series & Trend Analysis

When performance improved or declined, turning points and events, WQ trends and their effect on the shrimp, and the feed–growth relationship over the culture.

5.1 Water-quality explorer (pick parameter, toggle ponds)

Parameter:

Morning Dissolved Oxygen (ppm) ▾

Ponds:

● 2D407 · Control

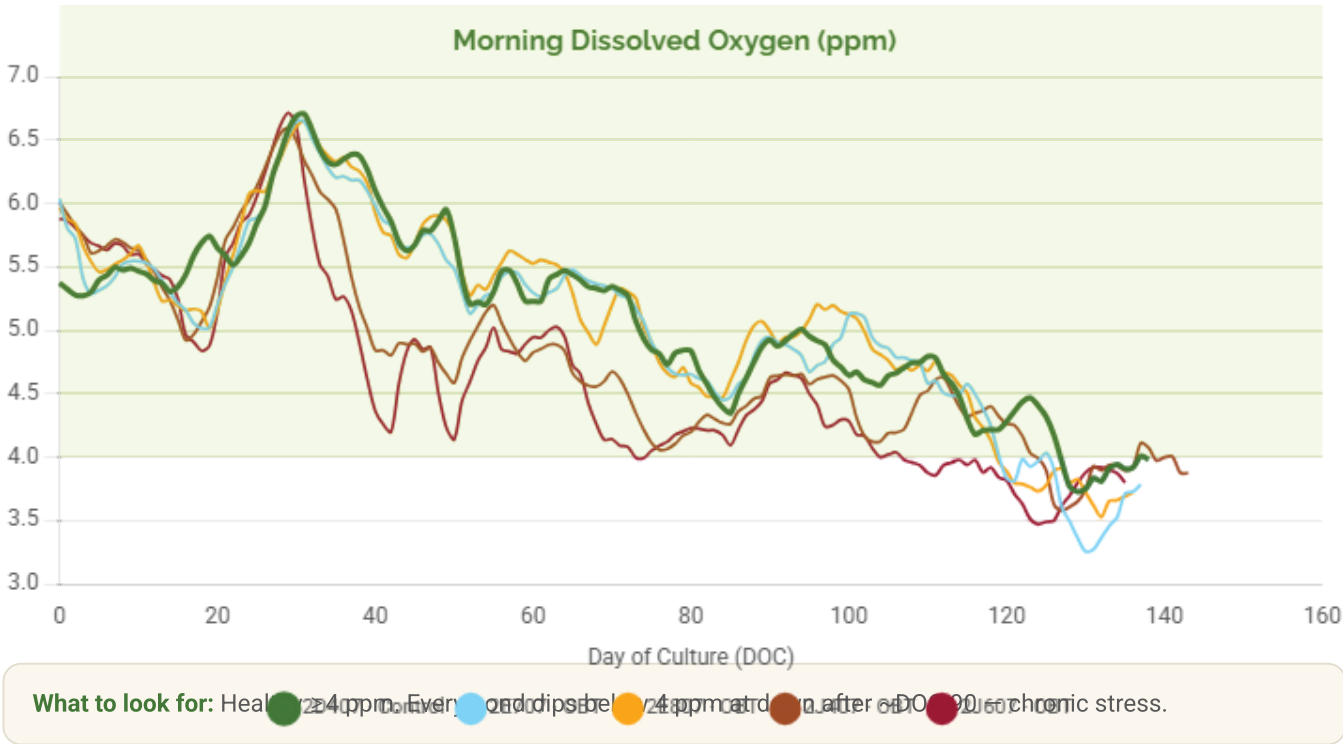
● 2E707 · OBT

● 2E807 · OBT

● 2J407 · OBT

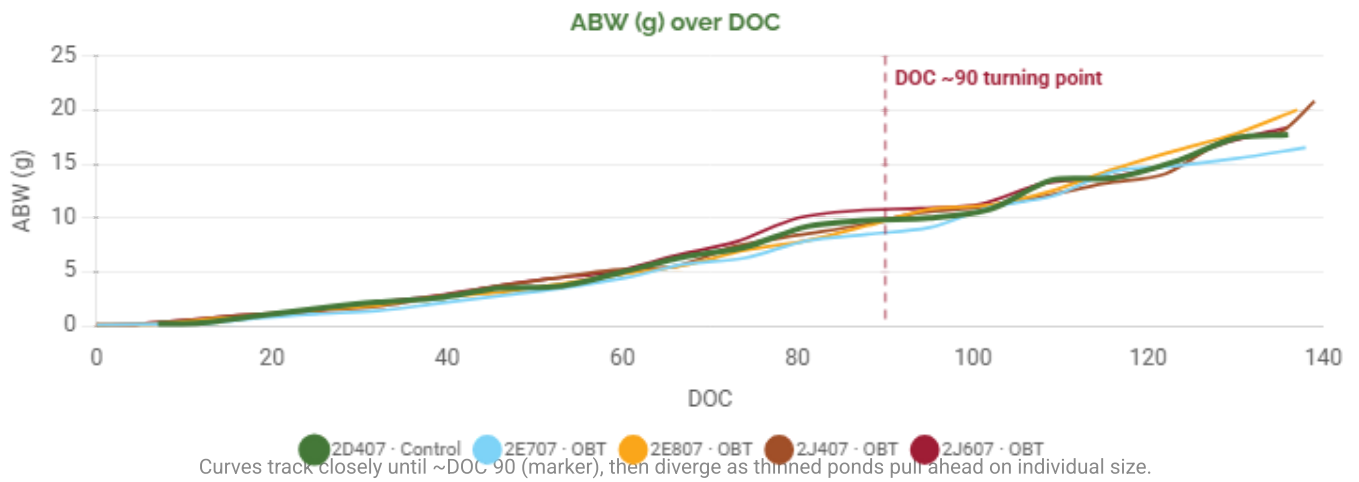
● 2J607 · OBT

✦ Ask AI — explain this chart



5.2 Growth curve with turning point

✦ Ask AI — explain this chart



5.3 Feed ↔ growth relationship

2D407

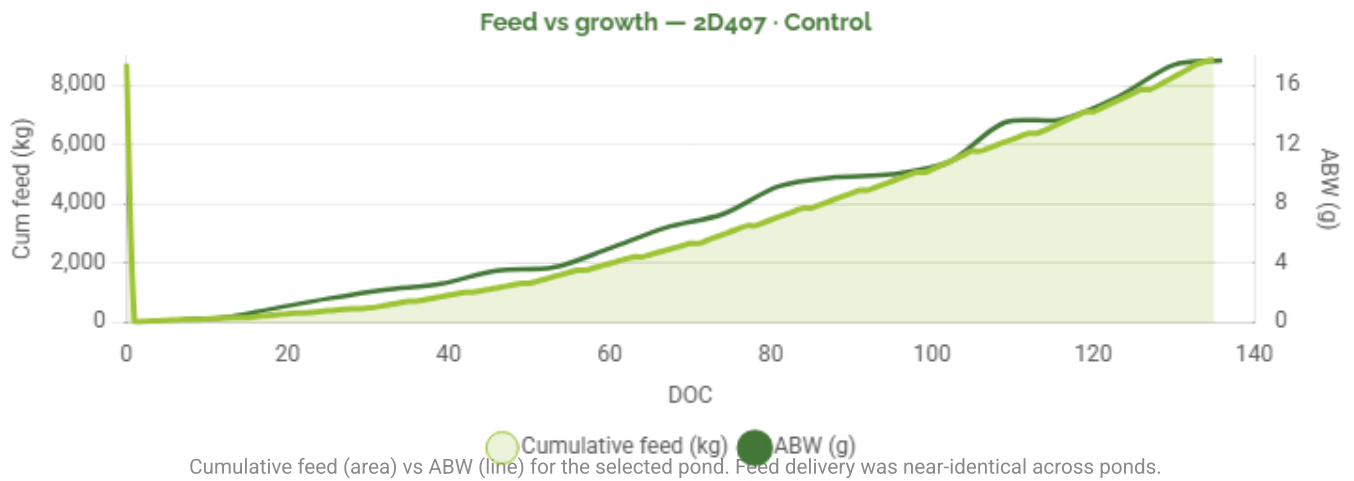
2E707

2E807

2J407

2J607

✦ Ask AI — explain this chart



5.4 Turning points & events timeline

- **DOC 0 (25/27 Dec):** stocking, 505,000 PL @ 0.1 g, ~20°C, salinity 40 ppt.
- **DOC 25–30:** early plankton bloom — morning DO peaks ~6.5 ppm; healthiest WQ window.
- **DOC ~90 onward:** morning DO begins sustained decline toward <4 ppm; growth curves diverge; rising biomass + seasonal warming (toward ~30°C) raise oxygen demand.
- **2J407 (timing varies):** ammonia/nitrite climb above peers — overload signature.
- **DOC 136–145 (May):** all five ponds harvested. 2J607 & control kept the highest survival; 2E807 & 2J407 were thinned but grew the largest shrimp.

WQ → performance effect: the late-culture dawn-DO squeeze is the clearest stress signal and coincides with peak biomass/feed load; pH, salinity and alkalinity stayed in comfort ranges and so did not constrain performance.

Statistical Analysis

Mean, median and standard deviation; % change; correlations; variability and anomaly detection; and the confidence/limitations of each finding.

6.1 Descriptive statistics — mean / median / SD

Parameter:

Morning DO (ppm) ▾

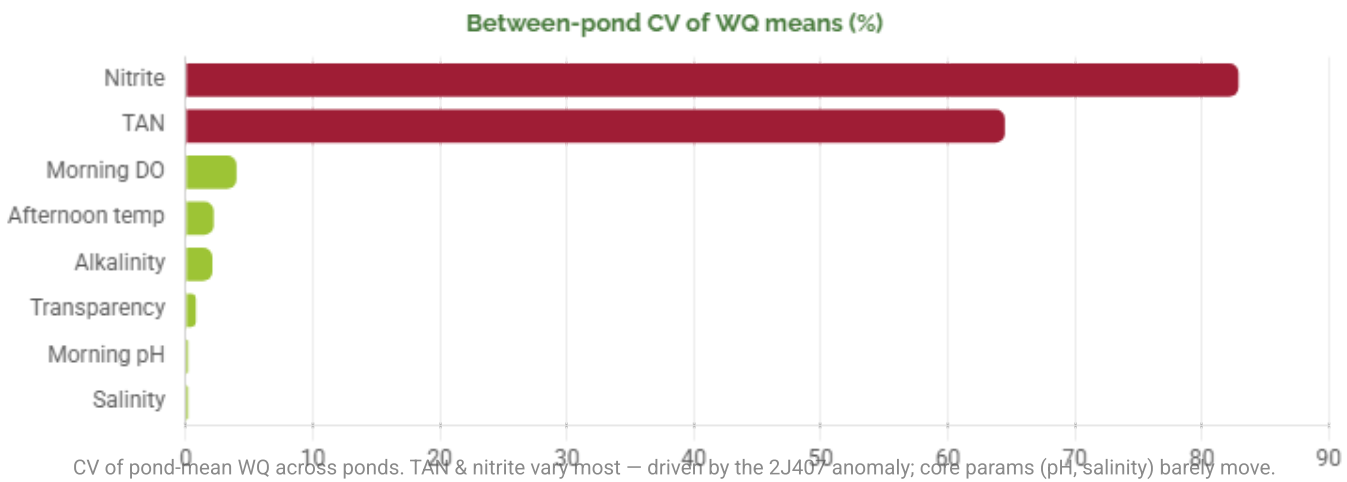
✦ Ask AI — explain this chart

Pond	Mean	Median	SD	Group
2D407	5.11	5.17	0.74	Control
2E707	5.07	5.10	0.80	OBT
2E807	5.13	5.11	0.80	OBT
2J407	4.78	4.68	0.75	OBT
2J607	4.65	4.33	0.83	OBT

Whole-cycle daily values per pond. SD shows day-to-day variability within a pond.

6.2 Between-pond variability & anomalies

✦ Ask AI — explain this chart



6.3 Within-pond DO trend (robust, n≈136/pond)

✦ Ask AI — explain this chart

Pond	Early DO	Late DO	Slope/day	r
2D407	5.28	3.92	-0.0145	-0.79
2E707	5.72	3.72	-0.0157	-0.78
2E807	5.84	3.66	-0.0156	-0.78
2J407	5.82	4.01	-0.0146	-0.80
2J607	5.81	3.90	-0.0165	-0.79

Linear regression of morning DO on DOC. All ponds decline, $|r| \approx 0.78\text{--}0.80$, $p \ll 0.01$ — the firmest statistical result.

6.4 Anomaly detection

- 2J407 TAN 0.19 / NO₂ 0.06 ppm — 3–6× the trial mean; clear outlier.
- 2E807 43.9% & 2J407 45.3% survival — the two lowest, ~13–15 pts below the best.
- Late-culture morning DO <4 ppm in every pond — threshold breach.
- 2J407 ABW 20.82 g & 2E807 19.96 g — high outliers, both explained by low survival (thinning).

6.5 Confidence & significance

- Treatment n=4, control n=1 — below replication for inferential testing; differences are **descriptive**.
- Cross-pond correlations (n=5) are hypothesis-generating, often confounded — not proof.
- Within-pond DO trend & the survival-drives-biomass decomposition are **statistically/arithmetically robust**.
- No randomisation metadata — seed-batch & location effects can't be separated from treatment.

SECTION 7

Hidden Insights

Patterns easy to miss, trade-offs, delayed effects and early-warning indicators — each as its own card.

PATTERN

Survival is the master variable

Survival varied more (CV 11.9%) than growth (8.4%). With identical stocking, ponds that protected survival (2J607, control) won on biomass, FCR *and* profit.

TRADE-OFF

A high ABW can be a red flag

The two biggest-shrimp ponds (2J407 20.82 g, 2E807 19.96 g) got there through heavy mortality thinning the pond — survivors had more food/space, yet both finished low on biomass. Always read ABW next to survival / standing density.

PATTERN

Profit hinges on kilograms, not cost

Total cost varied only ~2%; the entire ~\$5,500 best-to-worst profit gap was a yield story. Focus on yield (survival), not input shaving.

DELAYED EFFECT

Probiotic's only consistent signal is growth

ABW ran higher across treated ponds (+6.7% vs control), hinting at a digestion/gut benefit that shows up late — but it did not protect survival.

EARLY WARNING

A delayed, system-wide oxygen squeeze

Morning DO declines gradually and identically in every pond — easy to miss daily, yet all breach 4 ppm at dawn after ~DOC 90. Trigger pre-emptive aeration then.

EARLY WARNING

2J407 is a case study in overload

Ammonia/nitrite climbing while DO runs lowest is the signature of an overloaded pond — an actionable trigger to cut feed and raise exchange before a crash.

Report-Ready Conclusions

Each key conclusion as Finding → supporting data → interpretation → statistical support → recommendation.

C1 · The probiotic effect was inconsistent, not uniformly positive	
FINDING	On average treated ponds trailed control, but the single best pond was OBT.
DATA	OBT vs control: biomass −8.7%, survival −13.9%, FCR +11.1% worse; yet 2J607 was +4.7% biomass, +10% profit.
INTERPRETATION	High variance within the treatment — the product's effect is pond-dependent under current use.
STATISTICAL SUPPORT	Treatment mean below control on 3/4 KPIs; n=4 treated → descriptive, not significant.
RECOMMENDATION	Run a replicated randomised trial before any rollout decision.

C2 · Survival was the master variable	
FINDING	Survival drove biomass and FCR more than growth or routine WQ.
DATA	Survival CV 11.9% > ABW CV 8.4%; top-2 biomass = top-2 survival; best survival = best FCR.
INTERPRETATION	With identical stocking, retaining animals matters more than growing them bigger.
STATISTICAL SUPPORT	Biomass decomposition (survivors × ABW) + CV comparison; arithmetically robust.
RECOMMENDATION	Make early-survival protection the #1 operational KPI.

C3 · Late-culture dawn oxygen is the key environmental stressor

FINDING

Morning DO falls below 4 ppm in every pond after ~DOC 90.

DATA

~5.5 → 3.7–3.9 ppm; slope ~-0.015 ppm/day.

INTERPRETATION

Rising biomass + warming raise night O₂ demand; chronic dawn hypoxia suppresses feeding/immunity.

STATISTICAL SUPPORT

Within-pond regression $r \approx -0.78$ to -0.80 , $p \ll 0.01$, $n \approx 136/\text{pond}$ – robust.

RECOMMENDATION

Add dawn aeration from ~DOC 90; cap late feed/biomass.

C4 · 2J407 shows an actionable nitrogen early-warning

FINDING

Ammonia/nitrite outliers with lowest DO; harvested at just 45.3% survival.

DATA

TAN 0.19, NO₂ 0.06 ppm (3–6× peers); DO 4.81 ppm → biomass 4,763 kg, -22% profit.

INTERPRETATION

Classic overloaded-pond signature, now confirmed by a low-survival harvest; catchable early via periodic chemistry.

STATISTICAL SUPPORT

Dataset-maximum TAN/NO₂, consistent across tests; 2nd-lowest survival.

RECOMMENDATION

Set TAN>0.15 / NO₂>0.05 ppm action triggers to catch overload before it costs survival.

Prioritised actions

1

Don't roll out OBT on current data

Run a replicated, randomised trial (≥ 4 –5 ponds/arm, identical dosing).

Evidence: OBT mean below control on 3/4 KPIs; n too small.

HIGH

- 2** **Adopt 2J607 as the reference protocol** **HIGH**

Document its dosing & aeration management; replicate.
Evidence: Best biomass, FCR, survival; only +ve profit (+10%).
- 3** **Protect early-to-mid survival** **HIGH**

Review seed batches, stocking stress, biosecurity.
Evidence: Survival is master driver; 3 OBT ponds lost 20–25% more animals than control.
- 4** **Add dawn aeration from ~DOC go** **HIGH**

Cap late feed/biomass to hold morning DO >4 ppm.
Evidence: DO <4 ppm late in every pond ($r \approx -0.79$).
- 5** **Set hard WQ action triggers** **MEDIUM**

TAN>0.15 / NO₂>0.05 ppm → cut feed, raise exchange/aeration.
Evidence: 2J407's nitrogen excursion preceded its low-survival harvest (45.3%).
- 6** **Diagnose the thinned ponds** **MEDIUM**

Root-cause the mid-culture losses in 2E807 & 2J407 specifically.
Evidence: Both had top ABW but bottom survival → lost biomass & profit.
- 7** **Keep WQ buffers stable** **LOW**

Hold alkalinity ~135–145 ppm; monitor as a low-cost lever.
Evidence: Well-buffered ponds stayed stable all cycle; cheap to manage.
- 8** **Report ABW next to survival/density** **LOW**

Stops thinned ponds (high ABW, low survival) looking like winners.
Evidence: The 2E807 / 2J407 trap.